The information capacity of functional-equation windows

Samuel Lavery*

August 2026

Abstract

Fix an archimedean factor $G(s) = \prod_{i=1}^{m} \Gamma_{\mathbb{R}}(s+\lambda_i)$ with $0 < \lambda_1 \le \cdots \le \lambda_m$, its kernel $K(y) = \frac{1}{2\pi i}\int_{(c)} G(s)\, y^{-s}\, ds$, and a compact window $I = [a,b] \subset (0,\infty)$. The *windowed closure operator* sends a coefficient vector $(c_n)_{n\le N}$ to the reflection defect $x \mapsto \sum_n c_n\big(K(n/x) - \varepsilon\, x\, K(nx)\big)$ on I — the linearization of the question "which coefficient sequences satisfy this functional equation on this window?". We prove three quantitative laws for these operators and their character-twisted companions: exponential singular-value decay with an explicit Bernstein rate (so a single functional equation carries only $O(\log(1/\tau))$ constraint directions at relative precision τ, and the codimension of the windowed solution space is correspondingly small); the resulting linear-in-N deficiency and its depth-stratification; and a *burial law* for twists: twisting by a primitive character of conductor q, which rescales the kernel argument by $C = q^{(m_{\mathbb{C}}+\delta)/\cdots}$, does not destroy the channel's information but shifts its entire singular spectrum deeper by $\lambda_1 \log_{10} C$ decades, the minimal archimedean parameter acting as the burial rate. Numerical spectra for the symmetric-power stacks of the discrete series of weight 12 agree with all three laws, including the predicted 7.9-decade burial of the conductor-3 channel at the Sym^5 stack. We record the interpretation for converse theory: exact rigidity is a statement at infinite depth, per-channel information at any finite depth is logarithmic, and conductor growth throttles twist channels at a rate set by the archimedean type.

1 Setup

Throughout, $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(s/2)$ and $G(s) = \prod_{i=1}^{m}\Gamma_{\mathbb{R}}(s+\lambda_i)$ with real parameters $0 < \lambda_1 \le \cdots \le \lambda_m$. The associated kernel is

$$K(y) \;=\; \frac{1}{2\pi i}\int_{(c)} G(s)\, y^{-s}\, ds \qquad (c > 0),$$

a Meijer G-function of type $G^{m,0}_{0,m}$ in the variable $\pi^m y^2$ [1, Ch. V]; K is holomorphic on the sector $S_\phi = \{\, y :\ |\arg y| < \phi \,\}$ for any $\phi < m\pi/4$, positive on $(0,\infty)$, of polynomial growth y^{λ_1} as $y \to 0^+$ and of stretched-exponential decay $|K(y)| \le C_K \exp(-c_m\, |y|^{2/m})$ in any proper subsector [1, Ch. V, §5.7–5.10]. Fix a window $I = [a,b] \subset (0,\infty)$, a sign $\varepsilon \in \{\pm 1\}$, and define, for $N \in \mathbb{N}\cup\{\infty\}$, the *windowed closure operator*

$$A_N :\ \ell^2(\{1,\dots,N\}) \longrightarrow L^2(I), \qquad (A_N c)(x) = \sum_{n\le N} c_n f_n(x), \quad f_n(x) = K(n/x) - \varepsilon\, x\, K(nx).$$

*Numerical instruments cited as data: `tmp/adapter_sym_probe.py`, `tmp/adapter_rigidity_probe.py`, `tmp/adapter_twist_probe.py`, `tmp/adapter_twist_fast.py` in the helix–Frobenius repository. sam@sk-labs.com

Since sectors are invariant under dilations and inversion, each f_n extends holomorphically to any fixed Bernstein ellipse $E_\rho \subset S_\phi \cap \{\mathrm{Re} > 0\}$ for I (the image of $\{|w| = \rho\}$ under $w \mapsto \frac{b-a}{4}(w+w^{-1})+\frac{a+b}{2}$), and we set $M_n = \sup_{E_\rho} |f_n|$; the kernel decay gives $M_n \le C_\rho \exp(-c_\rho n^{2/m})$, so $\|M\|_2 < \infty$ and A_∞ is well defined and compact.

2 Window capacity

Theorem 2.1 (capacity of one functional equation)**.** *Let $\rho > 1$ be any Bernstein parameter with $E_\rho \subset S_\phi \cap \{\mathrm{Re} > 0\}$. Then the singular values of A_N (any $N \le \infty$) satisfy, for all $k \ge 0$,*

$$\sigma_{k+2}(A_N) \;\le\; \frac{2\sqrt{b-a}}{\rho - 1}\, \|M\|_2\, \rho^{-k}.$$

Consequently the relative τ-rank obeys

$$\#\{j : \;\sigma_j \ge \tau\, \sigma_1\} \;\le\; 2 + \log_\rho\!\Big(\frac{2\sqrt{b-a}\,\|M\|_2}{(\rho-1)\,\tau\,\sigma_1}\Big) \;=\; O\big(\log(1/\tau)\big).$$

Proof. Let $c \in \ell^2$, $\|c\|_2 \le 1$, and $g = A_N c$. By the Cauchy–Schwarz inequality g extends holomorphically to E_ρ with $\sup_{E_\rho} |g| \le \sum_n |c_n| M_n \le \|M\|_2$. Bernstein's theorem on polynomial approximation of functions analytic in an ellipse [2, Thm. 8.2] gives, for the space P_k of polynomials of degree $\le k$ (restricted to I, $\dim P_k = k+1$),

$$\mathrm{dist}_{C(I)}(g, P_k) \;\le\; \frac{2\,\sup_{E_\rho} |g|}{\rho - 1}\, \rho^{-k} \;\le\; \frac{2\|M\|_2}{\rho-1}\, \rho^{-k},$$

hence $\mathrm{dist}_{L^2(I)}(g, P_k) \le \sqrt{b-a}$ times the same bound. Thus the image of the unit ball of A_N lies within $\eta_k = \frac{2\sqrt{b-a}}{\rho-1}\|M\|_2 \rho^{-k}$ of the $(k{+}1)$-dimensional subspace $P_k \subset L^2(I)$, i.e. the $(k{+}2)$-nd approximation number of A_N is at most η_k; for compact operators between Hilbert spaces approximation numbers coincide with singular values [3, Thm. 2.5.3], giving the claim. The τ-rank bound is immediate. ∎

Corollary 2.2 (deficiency, stratified by depth)**.** *For every relative floor τ and every N,*

$$\mathrm{nullity}_\tau(A_N) \;\ge\; N - 2 - \log_\rho\!\Big(\frac{2\sqrt{b-a}\,\|M\|_2}{(\rho-1)\,\tau\,\sigma_1}\Big).$$

The deficiency of a single windowed functional equation grows linearly in the coefficient dimension, and the number of constraint directions grows only logarithmically in the depth $1/\tau$: the "hard band" at shallow depth is $O(1)$, and each further decade of depth releases at most $1/\log_{10}\rho$ additional directions.

Remark 2.3. This is the theorem behind two measured phenomena: the fixed shallow capacity of the bare functional equation across coefficient dimensions, and the *floor split* — per-channel yields of a few directions at 10^{-4} versus ~ 11 at 10^{-20} — which is the $\log(1/\tau)$-slope made visible.

3 The burial law for twists

For a primitive character χ of conductor q the twisted closure rows carry the rescaled kernel: with C the conductor scale of the twisted completed function (for the rank-d objects of §4, $C = q^{d/2}$),

$$f_n^{(q)}(x) \;=\; \chi(n)\big(K(n/(Cx)) - \varepsilon_\chi\, x\, K(nx/C)\big), \qquad A_N^{(q)} c = \sum_{n\le N} c_n f_n^{(q)}.$$

Theorem 3.1 (burial)**.** *Write the ascending exponent set of K at the origin as $\lambda_1 < \lambda' < \cdots$ (the poles of $G(s)$, i.e. the numbers $\lambda_i + 2k$, listed without multiplicity), let $r_0 = \#\{i : \lambda_i = \lambda_1\}$, and put $\Delta = \lambda' - \lambda_1 > 0$. Suppose $Nb \le C$ (every kernel argument on the window is ≤ 1). Then*

$$A_N^{(q)} \;=\; B_N \;+\; E_N, \qquad \operatorname{rank} B_N \le 2r_0, \qquad \|E_N\| \;\le\; C_1 \Big(\frac{Nb}{C}\Big)^{\lambda_1+\Delta} \sqrt{N},$$

where B_N is built from the leading Frobenius term $K(y) = \kappa_0 y^{\lambda_1}(1 + O(y^\Delta))$: its rows are $\chi(n)\kappa_0(n/C)^{\lambda_1}\big(x^{-\lambda_1} - \varepsilon_\chi x^{1+\lambda_1}\big)$ up to the r_0-fold multiplicity structure. Consequently the whole singular spectrum of the channel is suppressed by the factor $(1/C)^{\lambda_1}$ relative to an unstarved channel: on a logarithmic depth axis the channel's information is not destroyed but buried

$$\textit{burial depth} \;\approx\; \lambda_1 \log_{10} C \quad \textit{decades.}$$

Proof. The Mellin–Barnes representation gives the convergent Frobenius expansion of K at the origin by shifting the contour left across the poles of G [1, Ch. V]: $K(y) = \kappa_0 y^{\lambda_1} + O(y^{\lambda_1+\Delta})$ uniformly on $0 < y \le b$, with r_0 minimal-exponent terms when the minimal parameter has multiplicity r_0 (log terms, absorbed in the same rank count). Substituting $y = n/(Cx)$ and $y = nx/C$ and collecting the leading terms yields B_N, whose column space lies in the $\le 2r_0$-dimensional span of $x^{-\lambda_1}, x^{1+\lambda_1}$ (times logs); the remainder rows are bounded by $C_1(n/C)^{\lambda_1+\Delta}$ on I, and summing squares over $n \le N$ gives the stated norm bound. Weyl's inequality $\sigma_{k+2r_0}(A_N^{(q)}) \le \|E_N\|$ then confines all but $2r_0$ directions below the remainder scale, while the surviving leading block itself carries the prefactor $(1/C)^{\lambda_1}$; comparison with the unrescaled channel gives the depth shift. $\square$

Remark 3.2 (the archimedean type sets the burial rate). λ_1 is the minimal archimedean parameter of the stack. For the symmetric-power stacks of a weight-12 form, $\lambda_1 = 11/2$ at every odd rank; at Sym^5 (rank 6, $C = q^3$) the predicted burial of the conductor-3 channel is $\frac{11}{2}\log_{10} 27 \approx 7.9$ decades.

4 Data

All spectra are for the symmetric-power stacks of the discrete series of weight 12 on the window $I = [1, 1.8]$, with exact symmetric-power coefficients; floors are relative to σ_1.

Capacity and floor split. Bare functional equation: nullity 54/51/47 of 60 at rank 3 and 34/31/28 of 40 at rank 6 (floors $10^{-8}/10^{-14}/10^{-20}$); at matched shallow floors both ranks carry ~ 3 directions per channel, at 10^{-20} the rank-3 channels carry ~ 11 — the $\log(1/\tau)$ slope of Corollary 2.2.

Additivity at rank 3. Quadratic-twist channels $q = 3, 4, 5$ cut the rank-3 deficiency $47 \to 36 \to 24 \to 13$ (of 60, floor 10^{-20}): ~ 11 directions per channel, no saturation — consistent with the classical fact that GL(3)'s converse theorem needs GL(1) twists only.

Starvation and burial at rank 6. At $N = 120$ the shallow yields of $q = 3, 4, 5, 7, 8$ are $2, 2, 1, 0, 0$ — monotone in the conductor scale $C = q^3$, the shallow reading of Theorem 3.1. The deep spectrum of the $q = 3$ channel recovers ~ 10 directions at 10^{-20} while yielding ~ 2 at 10^{-4}: the channel's information sits ≈ 8 decades deep, matching the predicted $\lambda_1 \log_{10} C \approx 7.9$.

The completed five-channel deep curve, and the burial collapse. The full 30-digit spectra at $N = 120$ give per-channel deep yields (10^{-20} floor) of $10, 9, 7, 6, 4$ for $q = 3, 4, 5, 7, 8$. Assigning each channel its effective depth $20 - \lambda_1 \log_{10} C_q$ decades ($\lambda_1 = \frac{11}{2}$, $C_q = q^3$: effective depths 12.1, 10.1, 8.5, 6.1, 5.1), the yields collapse onto a single line of ≈ 0.85 directions per effective decade, constant across all five channels: within measurement resolution the decline is *entirely* burial — no intrinsic channel overlap is visible. Projected forward, only conductors with $\lambda_1 \cdot 3 \log_{10} q < 20$ (i.e. $q \lesssim 16$) contribute at this depth at all, for a total quadratic-twist ceiling of ≈ 98 of 120 directions: at rank 6, quadratic GL(1) twisting cannot reach rigidity at any fixed finite depth — while at rank 3, where $\lambda_1 = 1$ (the light zero-channel parameter), the same throttle is negligible to astronomical conductors, consistent with GL(3)'s GL(1)-only converse theorem. The projection is an extrapolation of the measured collapse, not a theorem.

5 Interpretation for converse theory

Three consequences, stated as interpretation rather than theorem. First, *exact* rigidity — the converse-theorem regime — is a statement at infinite depth: any finite-precision reading of a functional-equation system sees only logarithmically many constraints per channel, so the empirical hardness of converse problems is structural, not numerical accident. Second, rigidifying N coefficients at depth τ needs total capacity $\geq N$, hence #channels $\gtrsim N/\big(O(\log(1/\tau))\big)$ *before* conductor effects; twist families are not a technical convenience of the classical proofs but an information-theoretic necessity. Third, the burial law throttles high-rank twisting: the conductor scale $q^{d/2}$ grows with the rank d, so at fixed depth the usable twist channels thin out exponentially in the rank — a quantitative shape for why the twist requirements of converse theorems, and the difficulty of supplying them, escalate with d. Fourth, the mechanism suggests *why* higher-rank twists are what converse theorems demand: a GL(m) twist with archimedean parameters μ changes the tensor stack itself and can insert *light* channels (small $|\lambda_j - \mu|$-combinations) whose burial rate is near zero, opening unthrottled information channels that no GL(1) twist — which shifts conductors but never lightens the stack — can provide. On the carrier: a rail can only be read where its phasors have matured, and only a higher-rank twist can lower a rail's entry exponent. This is interpretation, testable by re-measuring with mock GL(2) Satake twists of tunable spectral parameter.

Remark 5.1 (geometric register). The objects of this note are the one-dimensional chart readout of carrier-native structure, and both laws state carrier facts. The variable x is the carrier's scale flow ($\log x$ the height along the axis), and $x \mapsto 1/x$ is the reflection through the origin exchanging the two conjugate ends of the double helix; the window is a height segment about the self-dual point, and the closure defect is the end-to-end *registration* of the helix against the anti-helix on that segment. Theorem 2.1 says a finite height window of registration has finite resolving power — logarithmically many fiber channels per decade of depth — because the carrier's clock is smooth; the Bernstein ellipse is the admissible tube around the axis, of width set by the rank's sector. Theorem 3.1 is the phasor entry law made quantitative: the exponent λ_1, a Mellin pole in the chart, is on the carrier the growth law of a phasor at birth (each phasor enters at height zero and grows continuously as y^{λ_1}), and a conductor-C rail's phasors are still in their entry regime at every height the base window sees — its information sits $\lambda_1 \log_{10} C$ decades deep because information lives where phasors have matured.

Remark 5.2 (scope and provenance). The operators studied here impose the functional equation *linearly*; no Euler-product or multiplicativity constraint is imposed, so nothing here bounds the classical converse theorems themselves, whose hypotheses live at exact depth with arithmetic structure. The nearest existing body of results known to us is the Selberg-class rigidity theory of Kaczorowski and Perelli (classification at small degree from functional equation plus Euler product); a systematic comparison is a stated obligation before any novelty claim for the framework. The numerical spectra are data, not proof; the theorems above are unconditional statements about the operators.

References

[1] Y. L. Luke, *The Special Functions and Their Approximations*, Vol. I, Academic Press, 1969.

[2] L. N. Trefethen, *Approximation Theory and Approximation Practice*, SIAM, 2013.

[3] A. Pinkus, *n-Widths in Approximation Theory*, Springer, 1985.

[4] J. Kaczorowski, A. Perelli, *On the structure of the Selberg class, I–VII*, Acta Math. **182** (1999) 207–241 and sequels.